STREAMING ADDENDUM

# naijaDNA Streaming Budget and Platform Recommend

This addendum supplements the corrected scaling report with streaming-specific budget guidance and a platform recommendation matched to the current codebase.

Date: 2026-04-07 Current stack: LiveKit already integrated in backend and Flutter

The current codebase already uses LiveKit for stream room tokens and media sessions. Keeping LiveKit is the lowest-change path. Switching to a broadcast-first platform like Mux becomes attractive only when your audience pattern shifts toward large passive viewing.

## Current Technical Fit

The backend already mints LiveKit participant tokens and the Flutter stream experience already connects to LiveKit rooms.

- LiveKit backend tokens
- LiveKit Flutter client
- Interactive host and viewer flow
- Lowest migration effort

## Streaming Budget by Platform

| Platform | Best fit | Entry budget | Growth budget | Recommendation |
|---|---|---|---|---|
| LiveKit Cloud | Interactive live sessions, host plus guests, viewer participation, low rewrite risk | about $0 for testing or about $50/month to start shipping | about $500/month for the Scale plan, plus usage above included quotas | Best immediate choice for this codebase |
| Mux | Large one-to-many broadcasts with mostly passive viewers | about $0 to start, with 100,000 monthly delivery minutes free | usage-based; 200,000 delivered minutes at 720p is about $80 in delivery after the free tier, plus live input at $0.025/minute | Best if you pivot to broadcast-first viewing at larger scale |
| Daily | Interactive video and audio calls with simple participant-minute pricing | about $0 for the first 10,000 minutes each month | about $0.004 per participant-minute at lower tiers, plus optional streaming or recording charges | Good product, but not the best fit when LiveKit is already integrated |
| Agora | Interactive streaming with larger enterprise networking needs | free for the first 10,000 minutes in some usage bands | pricing gets more complex fast; proxy-heavy interactive streaming can hit a $500 monthly minimum base fee | Not the first recommendation for this codebase |

## Simple Streaming Budget Scenarios

These are practical planning scenarios for the current product direction rather than exact invoice forecasts.

| Scenario | Rough monthly usage | Most suitable platform | Budget guidance |
|---|---|---|---|
| Small editorial live sessions | 1-4 live sessions per month, tens of viewers, interactive host presence | LiveKit | Keep the existing LiveKit integration. Expect the Build or Ship tier to be the right place to start. |
| Regular community streams | Weekly interactive shows with 50-200 concurrent viewers | LiveKit | Budget around LiveKit Ship or Scale depending on viewer concurrency and bandwidth profile. |
| Large passive breaking-news broadcasts | Hundreds or thousands of mostly passive viewers | Mux or a LiveKit plus Mux hybrid | Viewer delivery becomes the main cost, and a broadcast-first platform usually wins on cost efficiency. |

## Platform Recommendation

### Recommended now: Keep LiveKit

For the current codebase, LiveKit is the right platform to keep using for streaming because it is already integrated in both backend and Flutter; the current stream feature is interactive, not just passive broadcast; changing providers now would create avoidable backend and client rewrite work; and LiveKit already fits the current room-token architecture and viewer join flow.

### Recommended later: add Mux if streams become broadcast-heavy

If the product evolves into large one-to-many broadcast events, the best medium-term architecture is likely LiveKit for host, backstage, or guest coordination and Mux for large-scale viewer delivery, while keeping the rest of the application flow unchanged.

Recommendation in one line: keep LiveKit now, because it matches the existing implementation; only move to Mux or a LiveKit plus Mux hybrid when the audience becomes mostly passive and stream viewer counts materially outgrow the economics of interactive WebRTC delivery.

## Streaming Rollout Path

1. Continue with LiveKit for the current interactive streaming feature.
2. Measure actual monthly stream minutes, concurrency, and average viewer counts.
3. If streams stay interactive, scale within LiveKit.
4. If broadcasts become large and mostly passive, introduce Mux for viewer delivery while keeping LiveKit for host-side interactivity.

## Sources and Pricing References

These are the public pricing and planning references used for the streaming budget guidance in this addendum.


1. LiveKit pricing calculator and plans: https://livekit.io/pricing/calculator
2. LiveKit pricing guide for video conference and livestream: https://livekit.io/field-guides/guide/estimating-pricing-video-conference-livestream
3. LiveKit scale and WebRTC connection limits: https://livekit.io/field-guides/guide/plan-livekit-scale
4. Mux pricing: https://www.mux.com/pricing
5. Mux live input and delivery pricing details: https://www.mux.com/live and https://www.mux.com/docs/pricing/video
6. Daily pricing: https://www.daily.co/pricing/video-sdk/
7. Agora interactive live streaming pricing: https://www.agora.io/en/pricing/interactive-live-streaming/


### Practical Summary

If you want the lowest-risk next step, keep LiveKit and budget streaming as an extension of the current stack. If your future newsroom strategy shifts to large passive live events, treat Mux as the broadcast delivery option to evaluate next. That gives you a sensible progression: LiveKit now, LiveKit plus Mux later if audience shape demands it.